Hypotheses and measurable outcomes

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Record the expectation. Keep the denominator. Disclose the revision.

Define directional primary outcomes and distinguish observed patterns, secondary gains and exploratory changes.

Keep one habit: Outcome contract → ordered primary/revision ledger → ten-case picture → six-case coverage practice → preserved-plan audit → independent twelve-case metric.

MAIN LESSON · SECTION 1

Record the predicted outcome before inspecting results

> Keep the original primary question when a different outcome looks better.

Learning goals

I can explain the **mechanism** and terminology of hypotheses and measurable outcomes.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve measurable questions and fair comparison fields

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall A literature comparison matrix. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain hypotheses and measurable outcomes using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of hypotheses and measurable outcomes

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define the primary metric, direction and failure treatment

> A measurable expectation needs an outcome rule that another reviewer can apply.

Begin with support prediction and later speed discovery

Imagine predicting that version B will give more supported answers, then noticing it is faster. Speed may be useful, but changing the headline after seeing it does not make speed the original primary outcome. Keep the dated or ordered plan, its scoring rule and denominator, then label the revised claim exploratory. A testable prediction should also say which observed pattern would fail to meet the expectation.

Define a hypothesis and primary outcome

A **hypothesis** predicts a measurable comparison under stated conditions. Declare **primary outcome** and direction before inspecting results.

Complete fictional plan and outcome contract

Outcome-plan item	Complete fictional classroom rule
Main directional hypothesis	B supported-answer rate exceeds A on the same planned cases
Primary metric	Supported completed answers / all planned eligible cases
Support rubric	Substantive answer agrees with approved source under the fixed rubric; no unsupported factual claim
Failure treatment	Unsupported, abstention and timeout remain in denominator and do not count as supported answers
Secondary outcomes	Supplied mean duration, unsupported count, abstentions and timeouts, recorded separately
Post-result change	Speed headline chosen after inspecting results is exploratory; original primary remains visible
Separate picture	Ten reserved cases in the scenario; prediction B has fewer unsupported cases, defined by absent-from-card factual claims
Abstention caution	No unsupported factual claim does not automatically mean a supported useful answer; retain answer coverage
Scope	Local plan and constructed practice, not actual preregistration, reserved study, power calculation or significance claim

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace when the question changed as well as what it predicts

> A transparent revision is useful evidence; a hidden switch misstates the plan.

Original support hypothesis and outcome separation

Toy **hypothesis**: method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately.

Complete primary-versus-post-result ledger

Main ordered record	Meaning	Honest reporting
Before inspecting results	Primary is supported-answer improvement B>A	Record cases, versions, rubric and failure/denominator rule
After seeing faster responses	New information concerns latency	Report the observed speed result with its measurement conditions
Team switches its primary claim to speed	Outcome chosen after results	Label exploratory/post-result revision; retain original hypothesis
Support result not supplied by the short scenario	Cannot infer whether the original primary improved	Request/report it rather than replacing it

Main ordered record	Meaning	Honest reporting
Future confirmatory comparison	New plan and genuinely reserved eligible cases needed	Do not rebrand already revealed practice as unseen testing

State missing outcomes before deciding

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Hypotheses and measurable outcomes — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
After seeing faster responses, the team changes its primary claim from support improvement to speed improvement.	Toy hypothesis: method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately.	Report the change as exploratory/post-result revision. Preserve the original support hypothesis; do not present speed as the prespecified primary outcome.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict friendly wording with a tied unsupported count

> A different appealing quality cannot substitute for the recorded outcome.

Current task and response

- After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict friendlier B with same unsupported count

- Change: B sounds friendlier but has the same unsupported count as A.

▼ **Check after predicting**

Prediction: The stated improvement **hypothesis** is not supported on these cases.

Reason: Friendliness is a different outcome from the predefined unsupported-case count.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read primary outcome, appealing secondary result and coverage separately

> Keep the six-case added practice distinct from the picture's ten-case design.

Main post-result switch and checked correction

After seeing faster responses, the team changes its primary claim from support improvement to speed improvement.

Report the change as exploratory/post-result revision. Preserve the original support **hypothesis**; do not present speed as the prespecified **primary outcome**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate ten-case picture and honest outcome scope

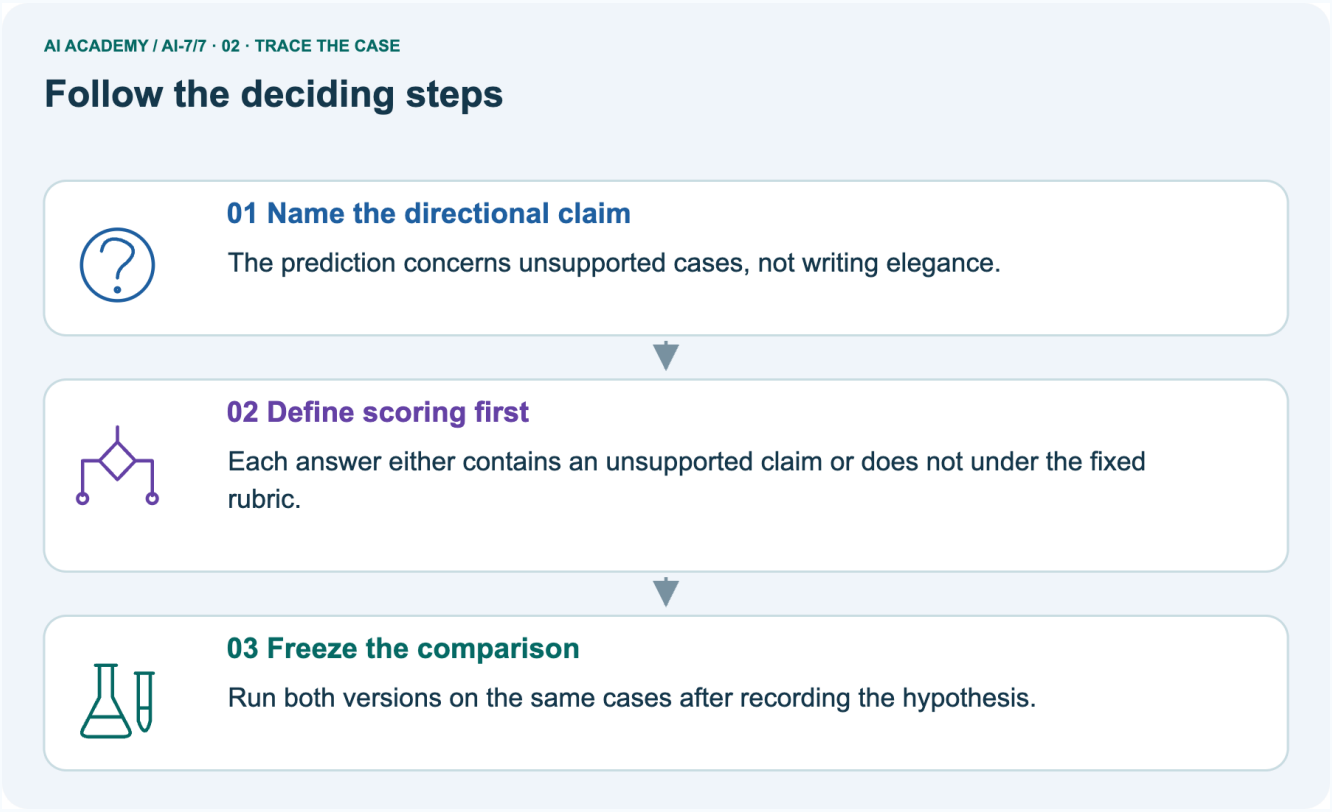
Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

Ten cases are reserved. Unsupported means a factual claim absent from the approved card. **Prediction**: B has fewer unsupported cases than A.

The picture describes a separate ten-case hypothesis: B should have fewer cases containing a factual claim absent from the approved card. Score that rule on the same cases after fixing the plan. In the changed scenario, B sounds friendlier but ties A's unsupported count; the strict descriptive reduction prediction is not met in that sample.

Added practice can illustrate a tie of two unsupported cases each, but that number is constructed, not an actual reserved-run result. Friendliness has no supplied primary rubric. Equal sample counts do not prove population equality or absence of any effect. Because abstention can reduce unsupported claims without answering, record answer coverage and abstentions too.

Trace direction, scoring and fixed comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name the directional claim	The prediction concerns unsupported cases, not writing elegance.
2. Define scoring first	Each answer either contains an unsupported claim or does not under the fixed rubric.
3. Freeze the comparison	Run both versions on the same cases after recording the hypothesis.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **hypothesis** makes an expected pattern explicit before seeing outcomes. The measured definition must be clear enough for another reviewer to apply.

Added complete six-case support/speed practice

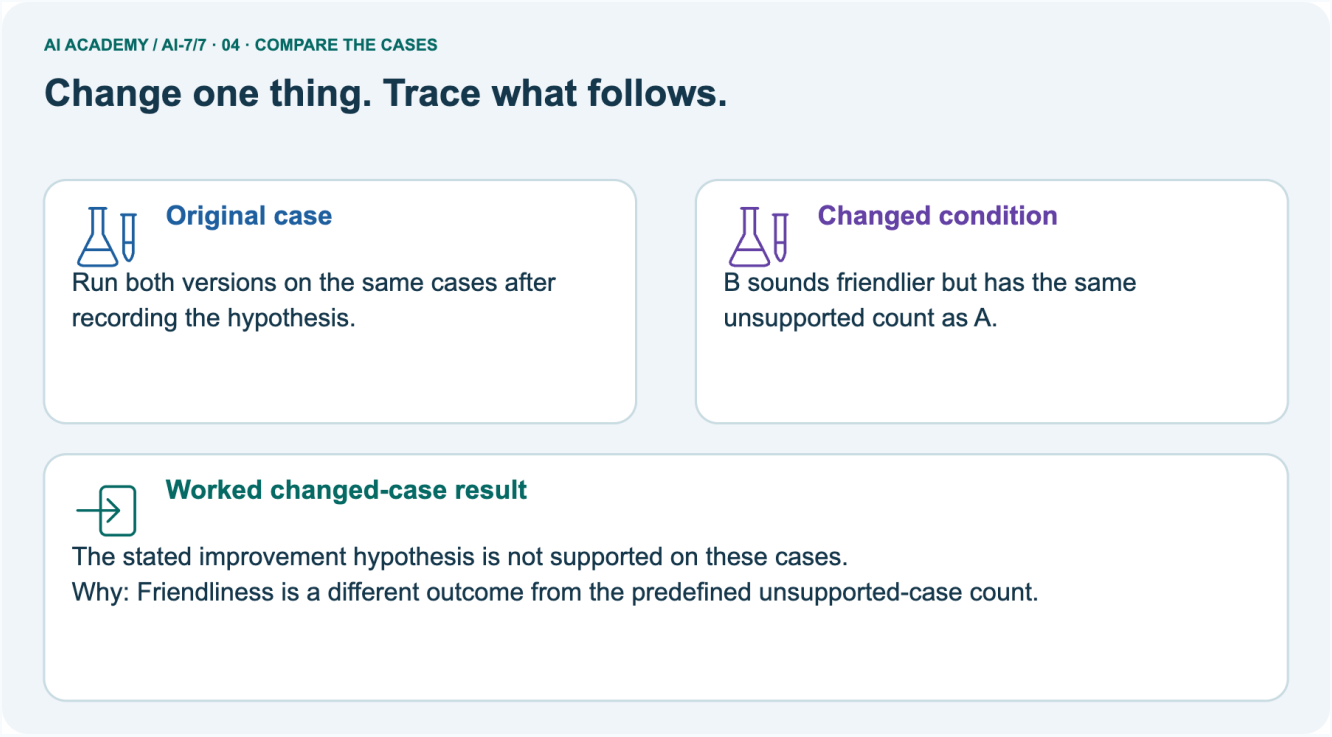
Added main practice: six paired cases	A	B	Primary interpretation
Supported completed answers	4	4	Both 4/6 =2/3; strict improvement not observed
Unsupported cases	1	0	Fewer B unsupported does not add a supported answer
Abstentions	0	1	B has one additional abstention

Added main practice: six paired cases	A	B	Primary interpretation
Timeouts	1	1	Keep in the fixed six-case denominator
Supplied mean final-outcome duration	3 seconds	2 seconds	B faster on this toy timing measure; separate from primary support
Evidence status	Constructed records	Constructed records	No actual model, participant or reserved experiment

Predict friendly wording with a tie

- Change one thing: B sounds friendlier but has the same unsupported count as A. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain power/equality limits and primary support

- No power calculation or significance claim is provided; a null descriptive difference is not proof of exact equality.

- Read the labelled picture: Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement **hypothesis** receive support?

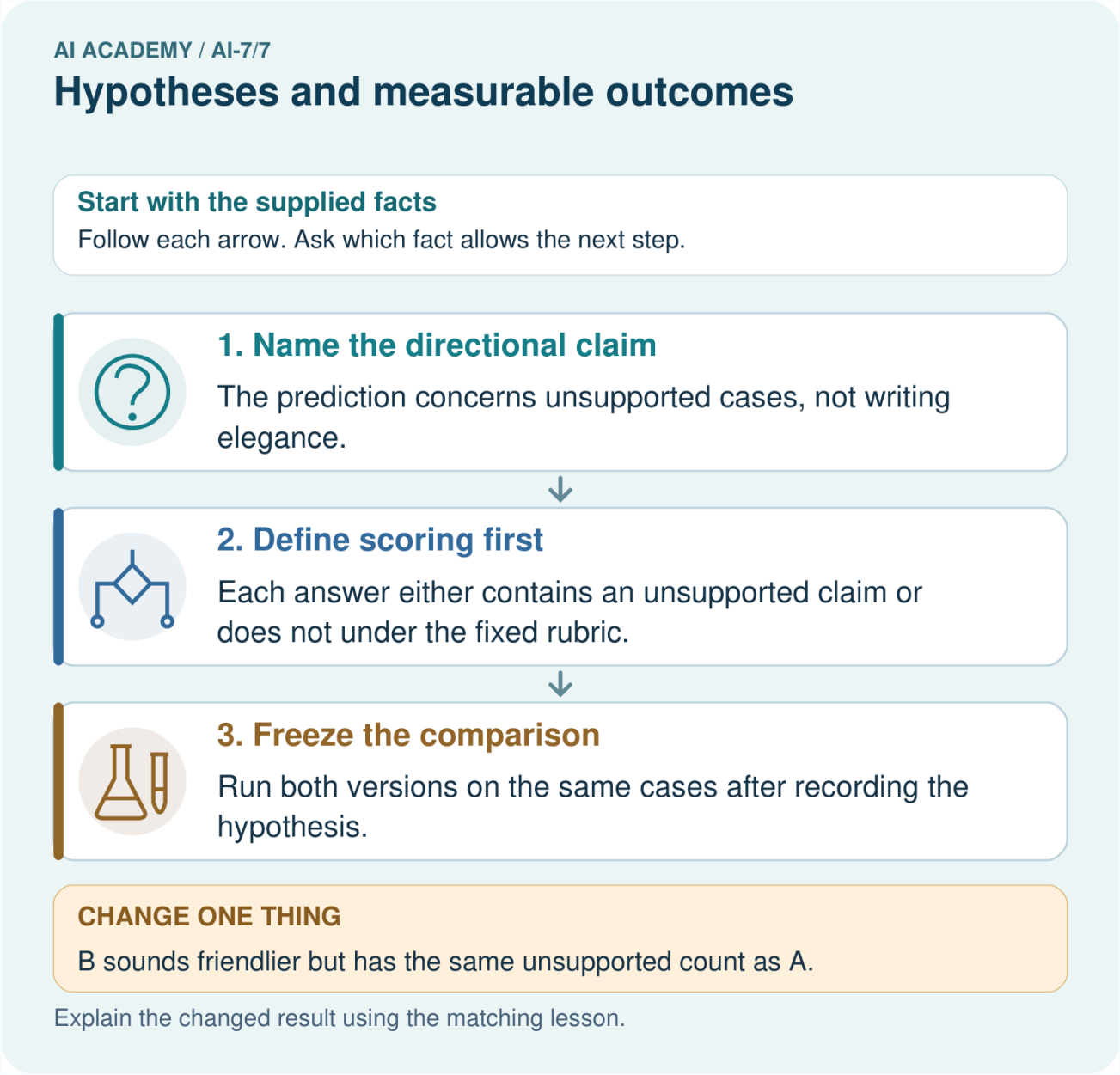
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Hypotheses and measurable outcomes

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit outcome switching, denominator changes and oversized inference

> Make revisions explicit instead of hiding the original question.

Core hypothesis qualification and failure analysis

A **hypothesis** can fail honestly; changing it after results does not erase the original test.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete metric, revision and coverage probes

Hypothesis probe	Expected interpretation
B strictly exceeds A on recorded primary	Observed sample pattern meets that directional expectation
B ties or falls below A	Strict improvement not observed in sample; no population-equality proof
B faster after primary support was fixed	Report speed separately; post-result new claim is exploratory
Unsupported count falls via abstention	May not improve supported-answer rate; report coverage
Timeout removed from denominator after results	Changes the declared rule; retain failure or disclose deviation
Friendlier wording without fixed score	Not evidence of the stated unsupported-count improvement
Plan appears before fixture evaluation in code	Ordered teaching example, not externally registered real study

Hypothesis probe	Expected interpretation
Known practice cases scored again	Regression/replay evidence, not unseen reserved evaluation
No power or statistical test supplied	Do not claim significance, equivalence or adequate precision

Guided repair and explanation

- A report omits this qualification: A **hypothesis** can fail honestly; changing it after results does not erase the original test. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise stating what would challenge the expected pattern

> Name the primary outcome, fixed denominator and limits.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example “Hypotheses and measurable outcomes”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Hypotheses and measurable outcomes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A’s unsupported count, does the recorded improvement **hypothesis** receive support?

> **My explanation using a pictured step and the given facts**

Reserve the selected twelve-case primary metric

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete toy plan, primary-result and revision record

> Check category counts and denominator before selecting a headline.

Current post-result outcome-switch investigation

After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. Show the intermediate steps and give the conclusion with its scope.

Runnable preserved-plan and coverage example

```
"""Declared toy analysis plan versus post-result changes; no actual
preregistration/study."""
from copy import deepcopy
from fractions import Fraction
from statistics import mean
from math import isfinite
VALID={'supported','unsupported','abstain','timeout'}

def summarize(rows):
 if type(rows) is not list or not rows:raise ValueError('nonempty complete-case fixture
required')
 ids=[];counts={state:0 for state in VALID};durations=[]
 for row in rows:
  if type(row) is not dict or set(row)!={'case_id','outcome','seconds'}:raise
ValueError('exact case fields required')
  if type(row['case_id']) is not str or not row['case_id'] or row['case_id'] in ids:raise
ValueError('unique IDs required')
  if row['outcome'] not in VALID:raise ValueError('declared outcome category required')
  if type(row['seconds']) not in (int,float) or not isfinite(row['seconds']) or
row['seconds']<0:
   raise ValueError('finite nonnegative supplied duration required')
  ids.append(row['case_id']);counts[row['outcome']]+=1;durations.append(row['seconds'])
 return
{'case_ids':ids,'counts':counts,'supported_rate':Fraction(counts['supported'],len(rows)),
         'mean_seconds':mean(durations),'denominator':len(rows)}
# Plan is recorded in code before evaluating these fully revealed practice fixtures.
plan={'primary':'supported_rate','direction':'B>A','failure_rule':'all planned cases stay
in denominator',
      'secondary':['mean_seconds','unsupported_count','abstention_count','timeout_count']}
saved_plan=deepcopy(plan)
def rows(outcomes,seconds):return [{'case_id':'Q'+str(i+1),'outcome':v,'seconds':seconds}
for i,v in enumerate(outcomes)]
```

```
a=summarize(rows(['supported']*4+['unsupported','timeout'],3))
b=summarize(rows(['supported']*4+['abstain','timeout'],2))
assert a['case_ids']==b['case_ids'] and
a['supported_rate']==b['supported_rate']==Fraction(2,3)
primary_pattern=b['supported_rate']>a['supported_rate']
assert not primary_pattern and b['mean_seconds']<a['mean_seconds']
revision={'stage':'post_results','proposed_new_claim':'B is faster','status':'exploratory
revision, not original primary'}
assert plan==saved_plan
# Separate pictured strict reduction question: supplied changed-case tie, no model is
scored.
picture={'reserved_case_count_in_scenario':10,'A_unsupported':2,'B_unsupported':2,
         'friendlier_wording':'qualitative scenario description, not a measured primary
metric'}
assert not picture['B_unsupported']<picture['A_unsupported']
print('Original declared toy plan:',plan)
print('Added practice A:',a,'B:',b,'strict primary improvement observed:',primary_pattern)
print('Post-result claim record:',revision,'original plan unchanged:',plan==saved_plan)
print('Separate pictured unsupported-count tie:',picture)
print('Abstention is not a supported answer. Fewer unsupported claims can coexist with
unchanged support and more abstention; retain all outcomes.')
print('This script defines/evaluates supplied fixtures in order, not a real
preregistration or reserved study. No significance, power, population equality, model
experiment or human scoring is established.')
```

Expected outcomes and study-status limits

1. Predict the added six-case supported rates, failure categories and speed summary before running or tracing. Both versions support 4/6, while B's supplied mean duration is2 seconds versus 3 and its unsupported case is replaced by abstention. The original primary plan remains unchanged and the later speed claim is explicitly exploratory.

   The separate picture illustrates a ten-case unsupported tie; no live scoring supplies it. The code orders a plan before scoring supplied practice fixtures, not a real preregistration or protected reserved experiment. No power calculation, significance result or population equality is established. Keep the independent twelve-case task separate.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a primary support prediction on twelve cases

> Specify metric and a sample result that fails the strict improvement expectation.

Complete selected independent task

Before running 12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Before running 12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

> **My independent procedure, result and limitation**
>
>
>
>
>
>

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the original hypothesis and every disclosed revision

> Keep primary, secondary and exploratory evidence visibly distinct.

Project task and artifact

Add a labelled hypotheses and measurable outcomes **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of hypotheses and measurable outcomes.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local scoring run or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the recorded prediction and the observed pattern without rewriting history

> Keep new questions useful and honestly labelled.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of hypotheses and measurable outcomes

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Hypothesis	A stated expectation to test
Primary outcome	The prespecified main measure
Exploratory result	An observation or analysis chosen after seeing evidence

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A **hypothesis** can fail honestly; changing it after results does not erase the original test. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Hypotheses and measurable outcomes — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
After seeing faster responses, the team changes its primary claim from support improvement to speed improvement.	Toy hypothesis: method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately.	Report the change as exploratory/post-result revision. Preserve the original support hypothesis; do not present speed as the prespecified primary outcome.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

By the end, I can	How I will show it
Explain the mechanism and terminology of hypotheses and measurable outcomes	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.
- Which labelled step matches this detail? Run both versions on the same cases after recording the **hypothesis**.
- 1. Name the directional claim
- 2. Freeze the comparison
- 3. Define scoring first
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Freeze the comparison Run both versions on the same cases after recording the **hypothesis**. A **hypothesis** makes an expected pattern explicit before seeing outcomes.

The measured definition must be clear enough for another reviewer to apply.

Changed-case question and optional hint

- Changed-case practice: B sounds friendlier but has the same unsupported count as A. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The stated improvement **hypothesis** is not supported on these cases.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The stated improvement **hypothesis** is not supported on these cases. Friendliness is a different outcome from the predefined unsupported-case count.

Repair and retain the record

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Review Methods results and limitations (ai-7/4). Goal: Explain the **mechanism** and terminology of methods results and limitations.

Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the result establish that B is best for outdoor farming?.

Return to this week and explain what you changed.

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

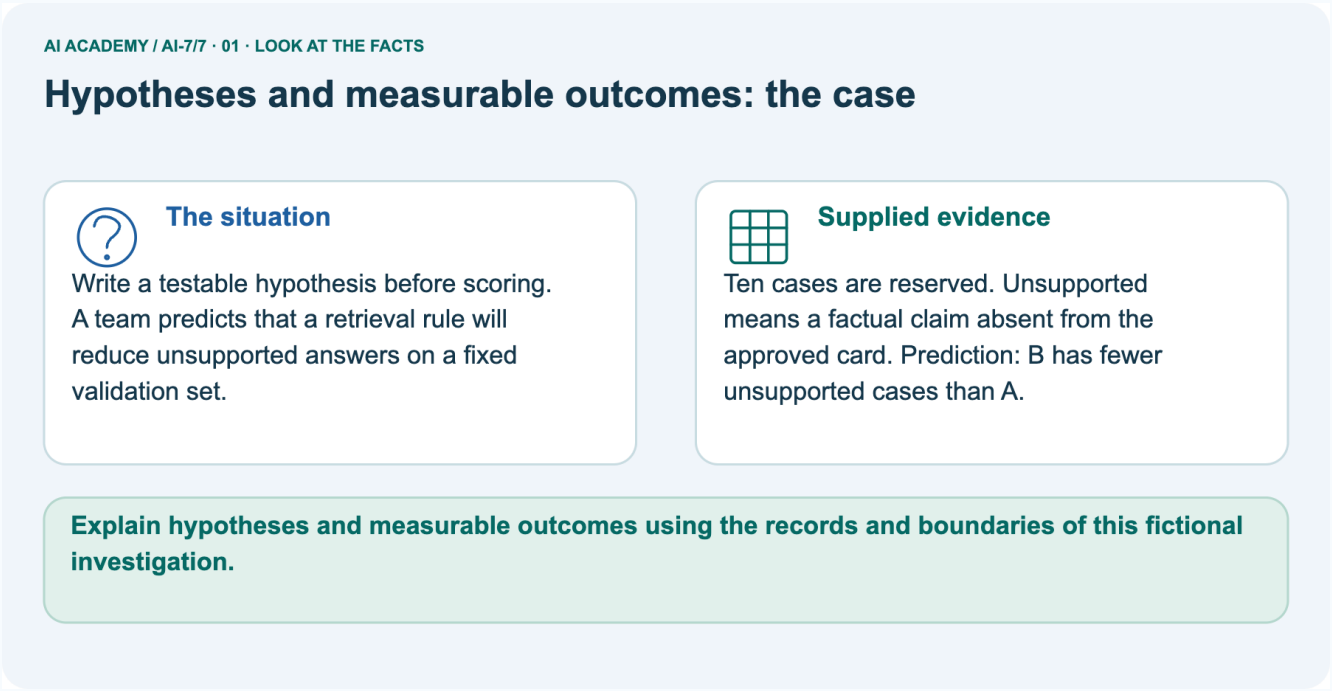
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Hypotheses and measurable outcomes: the case



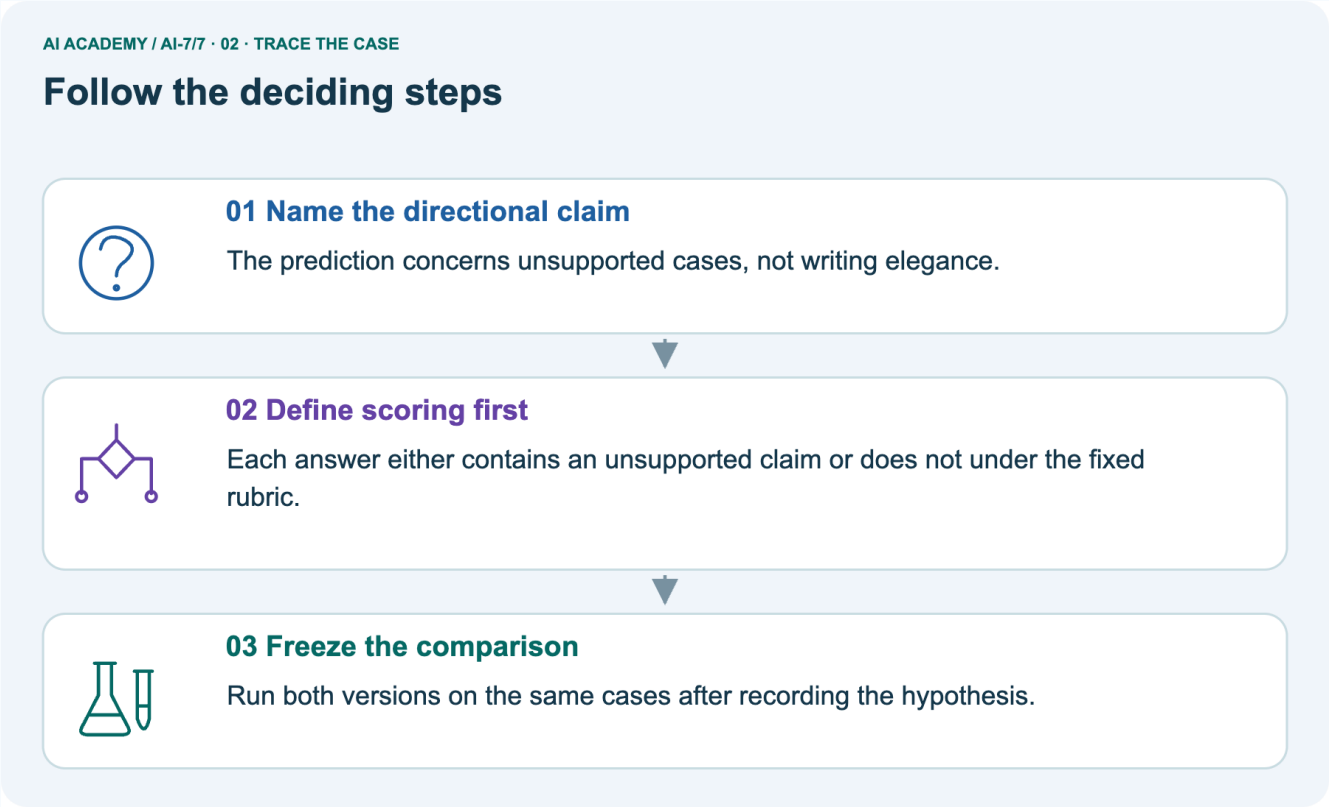
Follow the labelled picture with the matching explanation. Open the vector diagram

Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set. Supplied facts: Ten cases are reserved.

Unsupported means a factual claim absent from the approved card. **Prediction**: B has fewer unsupported cases than A.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

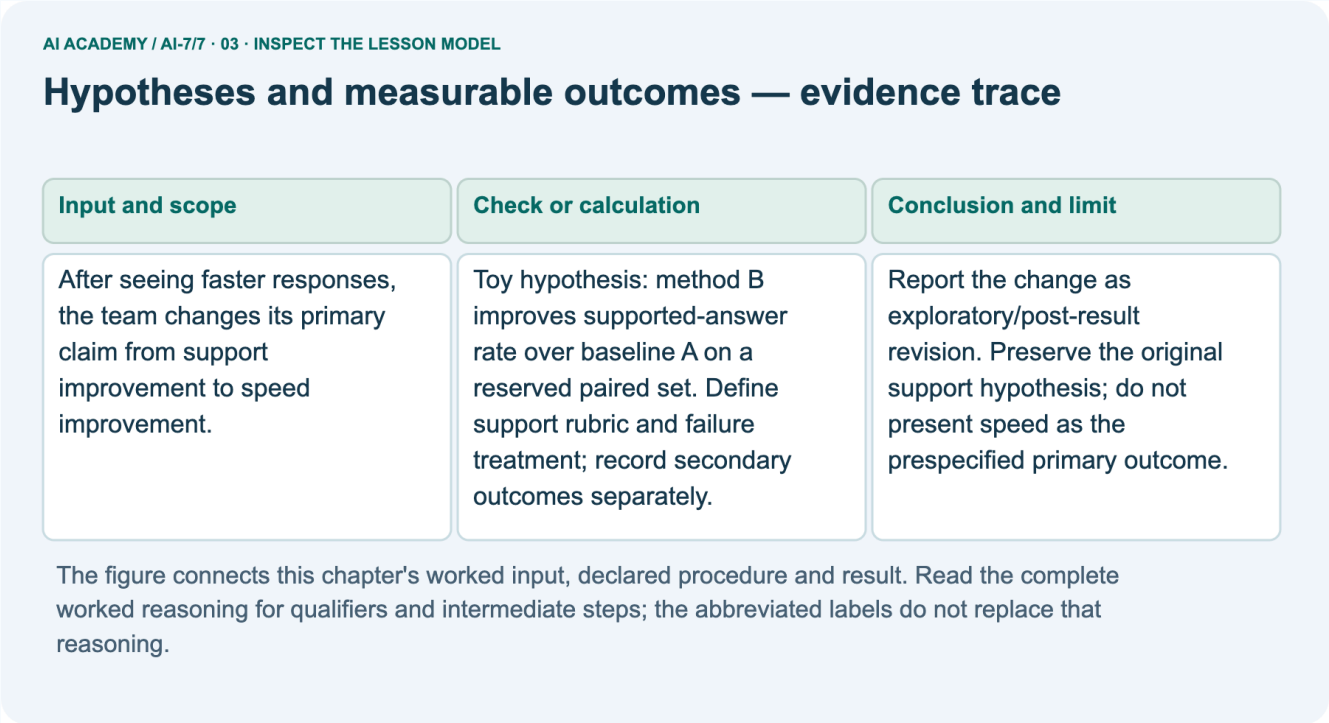


Follow the labelled picture with the matching explanation. Open the vector diagram

Name the directional claim: The **prediction** concerns unsupported cases, not writing elegance. → Define scoring first: Each answer either contains an unsupported claim or does not under the fixed rubric. → Freeze the comparison: Run both versions on the same cases after recording the **hypothesis**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Hypotheses and measurable outcomes — evidence trace

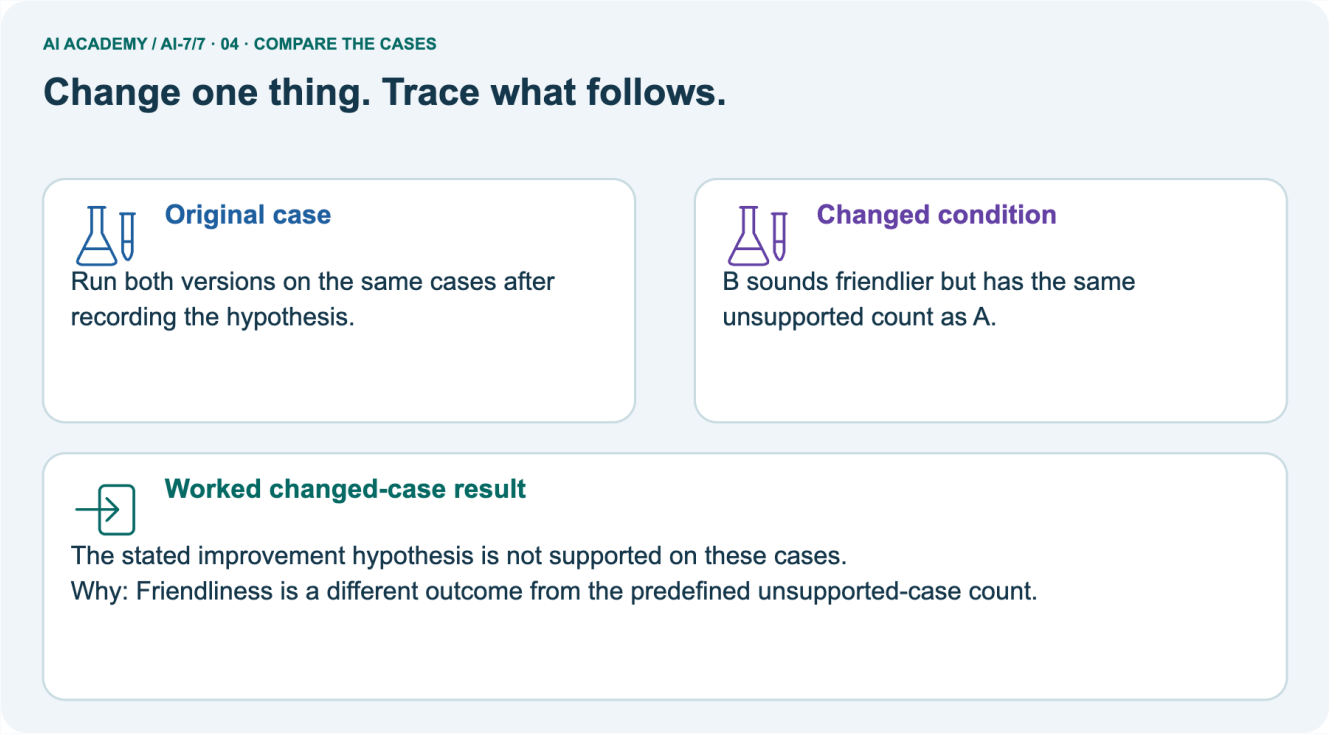


Follow the labelled picture with the matching explanation. Open the vector diagram

Hypotheses and measurable outcomes — **evidence** trace. After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. | Toy **hypothesis**: method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately. | Report the change as exploratory/post-result revision. Preserve the original support **hypothesis**; do not present speed as the prespecified **primary outcome**.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



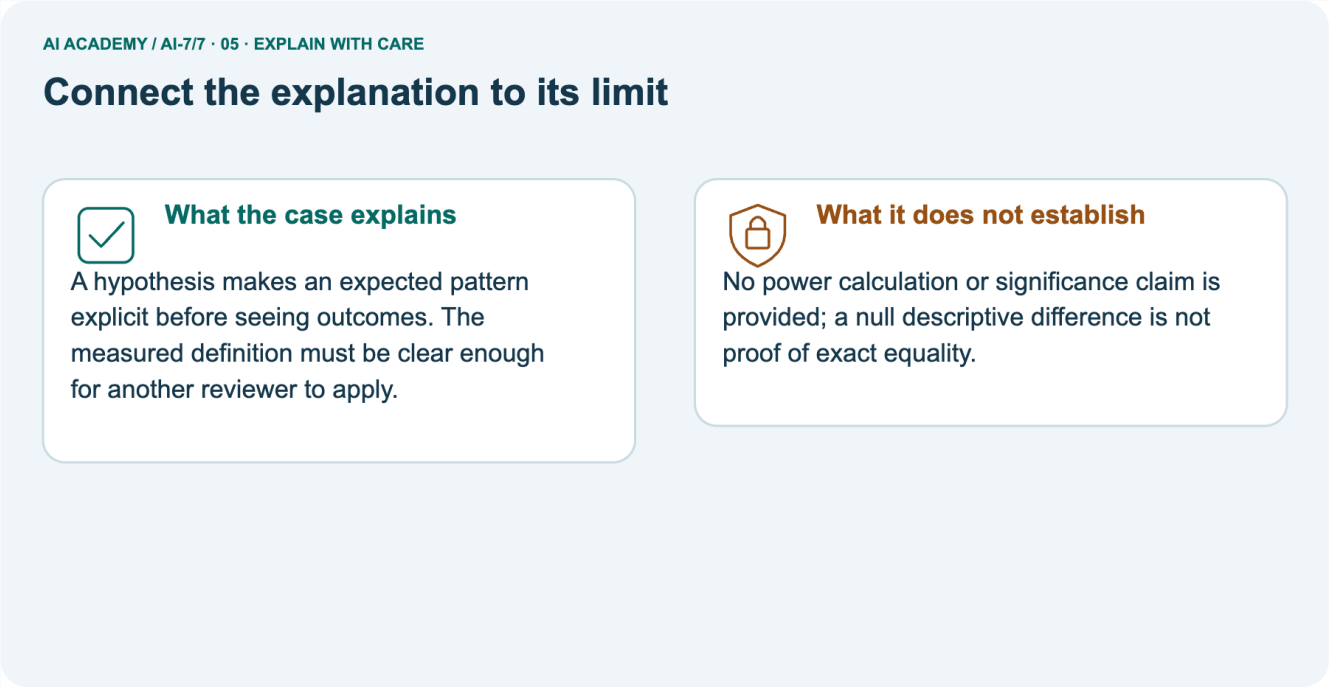
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Run both versions on the same cases after recording the **hypothesis**. Changed condition: B sounds friendlier but has the same unsupported count as A.

Expected result: The stated improvement **hypothesis** is not supported on these cases.
Reason: Friendliness is a different outcome from the predefined unsupported-case count.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



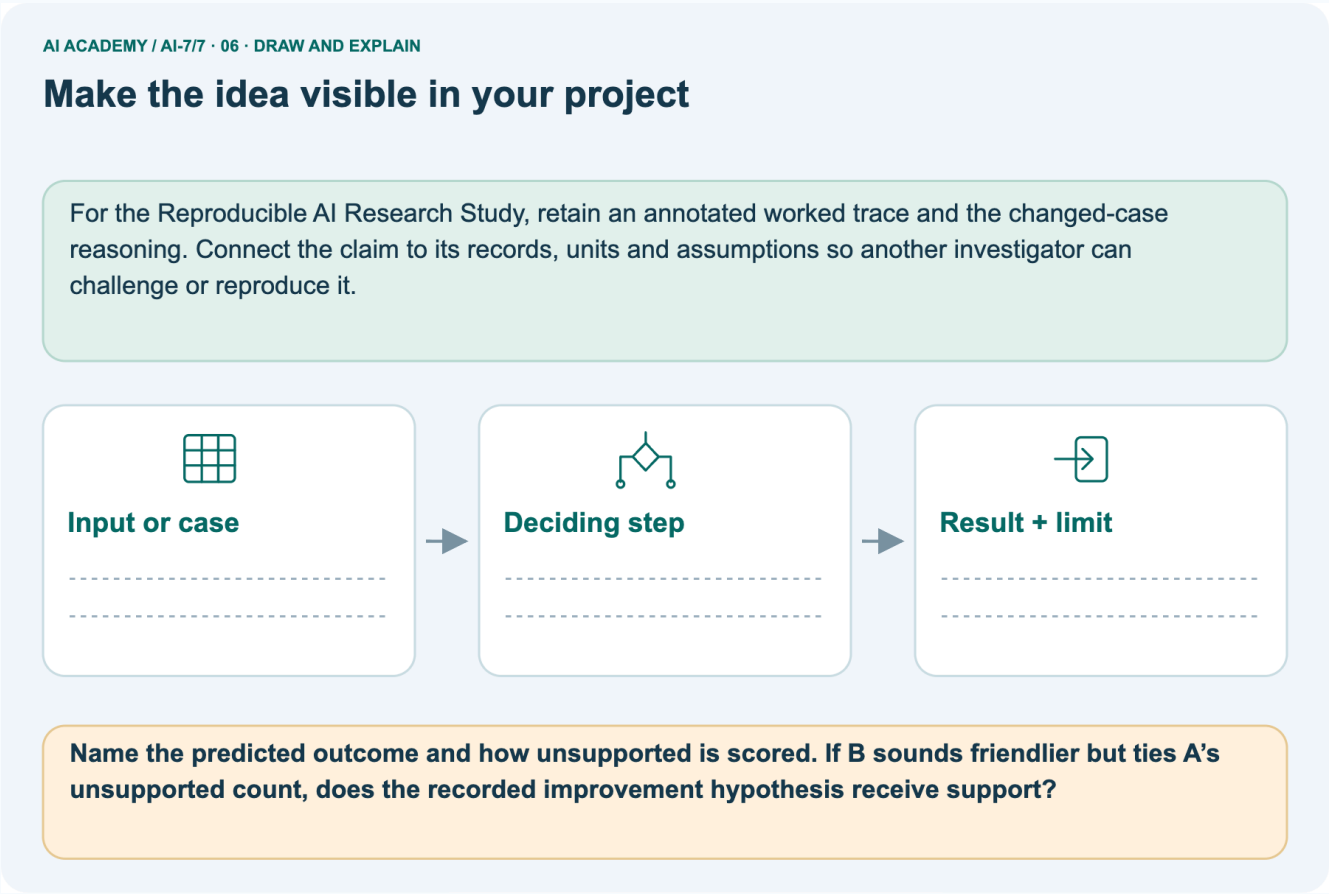
Follow the labelled picture with the matching explanation. Open the vector diagram

A **hypothesis** makes an expected pattern explicit before seeing outcomes. The measured definition must be clear enough for another reviewer to apply.

Limit: No power calculation or significance claim is provided; a null descriptive difference is not proof of exact equality.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement **hypothesis** receive support?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Hypotheses and measurable outcomes: the case

02. Follow the deciding steps

03. Hypotheses and measurable outcomes — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.